📜 IT Inventory System Review & Proposed Solutions

1. UI/Structure Issue: Branch Field Duplication

Issue

- “Branch” is shown per user entry instead of only once in the report header.

Root Cause

- Branch is stored/displayed at the record level instead of report-level metadata.

Proposed Solution

- Move **Branch field to report header level**.
- Remove Branch column from per-user rows.
- Ensure report query groups by selected branch.

Expected Outcome

- Cleaner report layout
- No redundant data
- Easier readability

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2. Reporting Logic Errors (Date Filtering & Status History)

Issue A: Date Filter Not Working Properly

Problem

- “From Date” is ignored.
- Results show all records up to “To Date”.

Root Cause

- Query likely uses:
- `WHERE date <= to_date`

  instead of:

  ```
  WHERE date BETWEEN from_date AND to_date
  ```

Proposed Solution

- Fix query logic:
- `WHERE date_acquired >= from_date`
- `  AND date_acquired <= to_date`
- Apply to ALL report types consistently.

Expected Outcome

- Accurate date filtering
- Reliable monthly reporting

Issue B: Status Automatically Changes Over Time

Problem

- "New" → automatically becomes "Functional" next month
- "Defective" → disappears next month

Root Cause

- Status is **time-dependent logic**, not **event-based tracking**.

Proposed Solution (IMPORTANT – Core Fix)

✅ Implement **Status History Tracking (Audit Trail Table)**

New Table: `asset_status_history`

Field	Description
asset_id	Asset reference
status	New / Functional / Defective / Repaired / Disposed
date_from	When status started
date_to	When status ended (nullable)
remarks	Optional

Logic:

- DO NOT overwrite status
- INSERT a new record on every status change

Reporting Fix:

To get status within a date range:

```
WHERE date_from <= to_date
AND (date_to IS NULL OR date_to >= from_date)
```

Expected Outcome

- You can view:
  - "New assets in January"
  - "Defective in March"
- Historical accuracy is preserved
- No disappearing records

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3. Employee Sorting Issue (Branch vs BLU)

Issue

- Branch and BLU employees are mixed alphabetically.

Proposed Solution

- Add **grouping + sorting logic**:

```
ORDER BY
  employee_type ASC,  -- Branch first, BLU second
  branch_name ASC,
  employee_name ASC
```

OR explicitly:

```
ORDER BY
  CASE
    WHEN employee_type = 'Branch' THEN 1
    WHEN employee_type = 'BLU' THEN 2
  END,
  branch_name,
  employee_name
```

Expected Output Format

```
[Branch Employees]
Jasaan Employee
Jasaan Employee
Jasaan Employee

[BLU Employees]
Manolo Employee
Manolo Employee
```

Expected Outcome

- Clean grouping

- Logical report flow

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4. Asset Naming & Data Structure Issues

Issue A: Long Asset Names in Reports

Problem

- Asset name includes:
- `Printer Single Function Printer OLIVETTI PR2+`

Root Cause

- Concatenation of Category + Subcategory + Model

Proposed Solution

- Separate fields clearly:
  - **Asset Name → Model only** (e.g., `OLIVETTI PR2+`)
  - Category/Subcategory → separate columns

Expected Outcome

- Clean printable reports
- Better formatting

Issue B: Brand & Model Redundancy

Problem

- Brand must be re-entered when adding a model
- Model dropdown shows unrelated brands

Root Cause

- No relational structure between Brand and Model

Proposed Solution

✅ Normalize database:

Tables:

- `brands`

- `models`
    - `model_id`
    - `brand_id` (FK)

Fix Dropdown Logic:

- When Brand is selected → filter models:

```
WHERE brand_id = selected_brand_id
```

Expected Outcome

- No duplicate brand input
- Cleaner data entry
- Correct dropdown filtering

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5. Missing Status Change Dates

Issue

- No date recorded when asset becomes:
    - Defective
    - Repaired
    - Disposed

Proposed Solution

- Use the **Status History Table** (same as Section 2)
- Require date input on every status change

Expected Outcome

- Full lifecycle tracking
- Audit-ready system

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6. Category & Subcategory Design Issue

Issue

- Subcategory exists in “Add New Asset” but causes redundancy and confusion.

Proposed Solution

- Move **Subcategory selection** to:

- `Asset Management → Equipment → Add New Equipment`
- Keep “Add New Asset” simplified

Enhance Data Capture:

Add structured fields per category:

For Printers:

- Type (Dot Matrix / Inkjet / Laser)
- Function (Single / Multi)
- Brand
- Model
- Connectivity

For CCTV:

- Type (Dome / Bullet / PTZ)
- Resolution
- Storage Type
- Brand
- Model

Expected Outcome

- Cleaner data structure
- Better filtering and reporting
- Less redundancy